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(12) **United States Plant Patent**
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(54) **THUJA PLANT NAMED ‘SILVANUS 1’**

(50) Latin Name: *Thuja occidentalis*
Varietal Denomination: **Silvanus 1**

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patent is extended or adjusted under 35
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(51) **Int. Cl.**
A01H 7/00 (2006.01)

(52) **U.S. Cl.** **Plt./213**

(58) **Field of Classification Search** **Plt./213**
See application file for complete search history.

(56) **References Cited**

PUBLICATIONS

UPOV–ROM GTITM, Plant Variety Database, 2006/01 GTI
Jouve Retrieval Software, Citation for Thuja ‘Silvanus 1. Sz.
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(57) **ABSTRACT**

‘Silvanus 1’ is a new variety of *Thuja occidentalis* that
exhibits medium dense branching and is suitable for use as
a solitary garden plant or for use as a field hedge.

2 Drawing Sheets

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Botanical classification: *Thuja occidentalis*.
Varietal denomination: ‘Silvanus 1’.

CROSS REFERENCE TO RELATED
APPLICATION

The present invention claims priority to Hungarian Plant
Patent Application No. F04 00057, filed on Sep. 22, 2004.

BACKGROUND OF THE INVENTION

The present invention comprises a new and distinct cul-
tivar of *Thuja occidentalis* known by the varietal name
‘Silvanus 1’. The new variety was discovered by the breeder
as a naturally-occurring branch (sport) mutation of *Thuja*
occidentalis ‘Smaragd’ in West Hungary. ‘Silvanus 1’ was
first asexually reproduced by semi-hardwood cuttings in
West Hungary. The new variety has been trial and field
tested and has been found to retain its distinctive character-
istics and remain true to type through successive propaga-
tions. Table 1 illustrates characteristics that distinguish ‘Sil-
vanus 1’ from its parent plant, ‘Smaragd’.

TABLE 1

Characteristic	‘Silvanus 1’	‘Smaragd’
Plant habit	Ovoid	Conic
Plant speed of growth	Slow	Medium
Color of upper leaf side in Summer	Light green	Green
Side branch position	Vertical	Horizontal
Scale leaf width	Narrow	Medium
Scale leaf thickness	Thin to medium	Medium

DESCRIPTION OF THE DRAWINGS

The accompanying photographic drawings illustrate the
new variety, with the color being as nearly true as is possible
with color illustrations of this type:

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FIG. 1 shows an entire plant of the new variety; and
FIG. 2 shows an entire field of the new variety.

DESCRIPTION OF THE PLANT

The following detailed description sets forth the charac-
teristics of the new cultivar. The data which defines these
characteristics were collected by asexual reproductions car-
ried out in West Hungary. The new variety was grown
outside without shade in average productivity soil after a
Summer season that experienced more rain than average.
‘Silvanus 1’ was four years old when described and was
grown free range (without a container). The color readings
were taken outdoors under natural light. Color references are
primarily to the R.H.S. Colour Chart of The Royal Horti-
cultural Society of London.

GROWTH AND PROPAGATION

Plant type and use: Squamifoliate evergreen plant (conifer).
It can be used solitarily in gardens or grouped as a field
hedge.
Growth habit: Ovoid.
Height: 50 cm.
Spread: 22 cm.
Hardiness: Zone 5, between 28.8–23.4° C.
Propagation protocol: The new variety grafts well and,
similar to ‘Smaragd’, ‘Silvanus 1’ can be propagated by
cuttings successfully. As the plant takes root well, the
cutting is the most important propagation protocol in the
nursery. The cuttings are taken immediately from the
semi-hardwood shoots taken off the mother plants. Please
note that only semi-hardwood cuttings can root, as soft
cuttings root off in the radication medium and cannot take
root. Also, fewer cuttings can be taken from ‘Silvanus 1’
because it is slower growing than ‘Smaragd’. The cuttings

are taken from the end of August until April, by the beginning of proliferation. The radication medium used is 0.2 mm granulation-sterile buddled gravel. For a better rooting, each cutting is dipped (about 1 mm deep) into 0.6% IBA (indole butric acid). The cuttings are grown in a temperate glasshouse. The temperature of the rooting medium should be between 19 and 20° C. and the temperature of the air must always be cooler. Low temperatures (up to -5° C.) also provide a stronger root system. Automatic overhead irrigation is supplied 2-3 times a day, depending on the weather. The cuttings take root in 9-12 weeks. The growing can then proceed in 9×9 plastic pots or in open ground.

Time to initiate roots: Approximately 42 to 56 days, with the temperature of radication medium being 19-20° C. and the maximum temperature of the air being 16-17° C.

Time to develop roots: Approximately 21 to 28 days, with the temperature of radication medium being 19-20° C. and the maximum temperature of the air being 16-17° C.

Rooting habit: Adventitious and fibrous root system.

Trunk:

Surface texture.—Reticulate and scaly.

Color.—Greyed-Orange group 177A.

Diameter.—28 mm at 50 mm in height.

Disease/pest resistance: No resistance or susceptibility has been observed to date.

STEM DESCRIPTION

Shape:

Twigs.—Upright.

Mature branches.—Upright.

Color:

Twigs.—Greyed-Orange Group 177B.

Young branches.—Greyed-Orange Group 164C.

Mature branches.—Greyed-Orange Group 177B.

Diameter:

Twigs.—8 mm.

Main branches.—5 mm.

Secondary branches.—2.5 mm.

Length (measured on a six-year old plant):

Twigs.—11.3 cm.

Main branches.—31.5 cm.

Secondary branches.—10.2 cm.

Surface description:

Twigs.—Fibrous, vertically striated.

Main branches.—Medium striation.

Secondary branches.—Minor striation.

Branching description: The branching is medium dense and it is arranged in vertical levels. The branching is similar to *Thuja orientalis* 'Aurea Nana' (syn: *Thuja orientalis* 'Berkmannii').

Branch aspect description: Similar to *Thuja occidentalis* 'Smaragd', but its structure is thinner and finer.

FOLIAGE DESCRIPTION

Leaf:

Color.—Green Group 143C.

Arrangement.—Opposite.

Fragrance.—Resin-smelling (if rubbed).

Shape.—Flattened squama.

Length.—2.5 mm.

Width.—Narrow, approximately 1.5-2 mm.

Division.—Overlap as a squama.

Base.—Oval.

Apex.—Pointed; sharp.

Margin.—Entire.

Surface texture.—Upper surface: Glandular points rise with an X motif because of overlapping scale leaves. Lower surface: X motif because of overlapping scale leaves.

Resin glands.—Location: At glandular point rises.

Shape: Square. Color: Green.

Cone description: Not available as cones are not yet present.

Seed description: Not available as seeds are not yet present.

I claim:

1. A new and distinct variety of *Thuja occidentalis* plant named 'Silvanus 1' as is herein described and illustrated.

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Fig. 1

Fig. 2

